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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

ETHICAL HACKING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Identify the main stages in the hacking methodology.	L1	CO1	**02**
	b.	Define OS fingerprinting	L1	CO2	**02**
	c.	List the common targets in social engineering attacks.	L1	CO3	**02**
	d.	State defensive strategies against DoS attacks.	L1	CO4	**02**
	e.	What is called web security?	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	illustrate the different types of threats and attacks.	L2	C1	**05**
		OR			
	b.	Discuss the steps involved in ethical hacking methodology.	L2	C1	**05**
3.	**a.**	Explain port scanning and its role in network security.	L2	C2	**05**
		OR			
	b.	Discuss the types of scanning techniques used in ethical hacking.	L1	C2	**05**
4.	**a.**	Explain sniffing and list common sniffing techniques	L1	C3	**05**
		OR			
	b.	Describe MAC flooding and ARP poisoning with example.	L2	C3	**05**
5.	**a.**	Explain DoS attack and how it differs from a DDoS attack	L2	C4	**05**
		OR			
	b.	Discuss the defensive strategies that protect against network level attacks	L2	C4	**05**
6.	**a.**	Illustrate the client-server relationship and its importance.	L2	C5	**05**
		OR			
	b.	Analyze the common vulnerabilities found in web servers.	L4	C5	**05**
7.	**a.**	Elaborate the various attack methods used against web applications.	L2	C5	**05**
		OR			
	b.	Illustrate the main security risks associated with insecure login systems.	L2	C5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Explain the need for information security in today's digital landscape and how it protects sensitive data in organizations from potential threats.	L2	C1	**10**

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		OR			
	b.	Explain in detail penetration testing with example.	L2	C1	**10**
9.	**a.**	Describe the elements of the CIA triad and how they form the foundation of information security.	L2	C1	**10**
		OR			
	b.	Interpret the role of risk management in protecting information assets.	L3	C1	**10**
10.	**a.**	Explain the different types of scanning used in ethical hacking.	L2	C2	**10**
		OR			
	b.	Analyze the process of OS fingerprinting and explain the various countermeasure.	L2	C2	**10**
11.	**a.**	Explain the concept of sniffing in network security.	L2	C3	**10**
		OR			
	b.	Illustrate how organizations can detect network level attacks.	L3	C3	**10**
12.	**a.**	Explain the following **i.** Session Hijacking **ii.** Brute force attack **iii.** Spoofing	L2	C4	**10**
		OR			
	b.	Discuss the countermeasures and defense strategies of web application attacks.	L3	C4	**10**
13.	**a.**	Explain the following **i.** Report writing and mitigation **ii.** Testing the web applications	L2	C5	**10**
		OR			
	b.	Analyze the common vulnerabilities found in web servers and applications.	L2	C5	**10**